

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name	Saher Fatima
Program/Batch	Cybersecurity Internship Program (B082)_Networkwalks
Date:	17 September 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) & W2-PM5(Zenmap Scanning)
Client/Target	- networkwalks.com (Footprinting Target) - Local VirtualBox NAT Network Subnet 10.0.0.0/24 (Scanning Target)
Permission Secured from Client?	Yes
Phases Covered	- Phase 1: Reconnaissance & Footprinting - Phase 2: Scanning & Network Discovery - Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors, and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorized access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the `networkwalks.com` domain using multiple Kali Linux tools (W2-PM1) and scanning my local virtualized environment with Zenmap (W2-PM5). One module covers the footprinting phase, while the other covers the scanning phase. Together, they demonstrate how a security professional moves from gathering public OSINT data to mapping active hosts on an internal network segment. This forms the Week 2 submission for my ongoing internship program at Networkwalks.

All commands were executed using Kali Linux 2025 in a virtualized environment. Every section includes the exact commands, actual outputs observed, space for evidence screenshots, and technical notes on why the findings matter from a defensive and offensive perspective.

3. Tools Used

Tool	Purpose
Kali Linux 2025	Operating system platform used for reconnaissance and scanning
WHOIS	Query domain registration details, registrar, creation dates, and name servers
WhatWeb	Fingerprint web application technology, web servers, CMS version, and plugins
Nslookup	Query DNS servers to resolve domain names to IP addresses
Curl (curl -I)	Inspect HTTP response headers, web server software, and security cookies
Wafw00f	Identify Web Application Firewalls (WAF) protecting the target
DNSRecon	Enumerate DNS records including SOA, NS, MX, A, TXT, and SRV records
Zenmap (Nmap GUI)	Graphically scan subnets, discover live hosts, and generate network topology maps

4. Activities Performed

4.1 Footprinting & Reconnaissance

Reconnaissance was conducted against `networkwalks.com` using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, and DNSRecon.

- **WHOIS:** Executed to retrieve public registration metadata and identify authoritative primary name servers (`ns6135.hostgator.com`, `ns6136.hostgator.com`).
- **WhatWeb:** Analyzed technology stack. Identified WordPress 7.1, WordPress Download Manager 3.3.58, Apache HTTP Server, Bootstrap 7.1, jQuery 3.7.1, and primary server IP `192.232.216.135`.
- **Nslookup:** Resolved `networkwalks.com` via Google Public DNS (`8.8.8.8#53`) to target IPv4 address `192.232.216.135`.

- **Curl** (`curl -I [https://networkwalks.com] (https://networkwalks.com)`): Retrieved HTTP/2 200 headers. Identified Apache server, `__wpm_client` session cookies, Nginx WordPress caching, and REST API endpoint `/wp-json/`.
- **Wafw00f**: Executed fingerprinting and detected **ModSecurity (SpiderLabs)** Web Application Firewall actively protecting the web server.
- **DNSRecon**: Conducted general DNS enumeration. Enumerated HostGator Name Servers, Mail Server (`mail.networkwalks.com` at `192.232.216.135`), SPF record (`v=spf1 +a +mx +ip4:50.87.144.87`), and 8 active SRV autodiscover records.

4.2 Network Scanning with Zenmap

For the network discovery activity, Zenmap was used inside Kali Linux to map active hosts on the local subnet (`10.0.0.0/24`).

- **Subnet & Scan Type**: Target subnet `10.0.0.0/24` with **Ping Scan** profile
(`nmap -sn 10.0.0.0/24`).
- **Live Hosts Identified**: 3 active hosts discovered in 2.20 seconds:
 1. `10.0.0.1` (QEMU Virtual NIC)
 2. `10.0.0.2` (VirtualBox Virtual NIC)
 3. `10.0.0.3` (Oracle VirtualBox Virtual NIC)
- **Topology Generation**: Generated interactive network topology map showing active host linkages and saved visual evidence for report documentation.

5. Risk Analysis / Impact

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Detailed CMS & Plugin Fingerprinting	WhatWeb exposed WordPress 7.1 and WP Download Manager 3.3.58	Attackers can search CVE databases for known public exploits targeting exact versions	Medium
2	Direct Public Server IP Exposure	Nslookup resolved domain directly to <code>192.232.216.135</code>	Reveals server physical host, allowing potential direct IP-based attacks bypassing DNS	Low
3	Technical Header Information Exposure	Curl exposed Apache server headers and REST API endpoints (<code>/wp-json/</code>)	Assists attackers in user enumeration and structural API path discovering	Low
4	WAF Fingerprint Identified	Wafw00f confirmed ModSecurity WAF in front	Informs attackers of security boundaries, helping them tailor	Low

		of web app	specialized WAF bypasses	
5	Comprehensive DNS Data Exposure	DNSRecon revealed internal cPanel mail records, SPF, and SRV endpoints	Exposes secondary target surfaces for phishing, mail-server targeting, or subdomains	Medium
6	Active Host Visibility on Local Subnet	Zenmap Ping Scan discovered 3 live active internal network hosts	Unauthorized devices or unpatched endpoints on internal subnets can be enumerated	Medium

Risk Level Key: Critical | High | Medium | Low

Note: These findings represent intelligence gathered during non-intrusive footprinting and host discovery phases. No actual exploitation or vulnerability verification was performed.

6. Recommendations

1. **Minimize Header Information:** Configure Apache and WordPress settings to suppress detailed server tokens and version numbers in HTTP response headers.
2. **Restrict REST API Endpoints:** Restrict unauthenticated public access to sensitive `/wp-json/` REST API endpoints where appropriate.
3. **Regular Software Patching:** Maintain strict updates for WordPress core, plugins, and web host modules to mitigate version-specific security risks.
4. **WAF Rule Maintenance:** Keep ModSecurity rules updated and properly tuned to filter automated reconnaissance and payload attempts.
5. **DNS Hygiene:** Audit public DNS and SRV records periodically to ensure legacy or unnecessary internal subdomains are not publicly exposed.
6. **Internal Subnet Monitoring:** Continuously monitor internal network segments using tools like Nmap to identify rogue or unknown connected devices.

7. Conclusion

During Week 2 of my Cybersecurity Internship, I successfully executed footprinting and network scanning methodologies. Using Kali Linux tools, I collected domain records, web stack details, DNS entries, and firewall signatures for `networkwalks.com`. Using Zenmap, I performed host discovery to detect live IP nodes and render network topology diagrams.

This exercise highlighted that thorough reconnaissance provides a detailed blueprint of a target's infrastructure before any active testing begins. Accurately documenting findings, assessing potential impact, and providing actionable security recommendations are vital responsibilities of a cybersecurity analyst.

8. Evidences Collected

kali-linux-2025.1c-virtualbox-amd64 (Fresh setup - 10.0.0.2) [Running] - Oracle VirtualBox

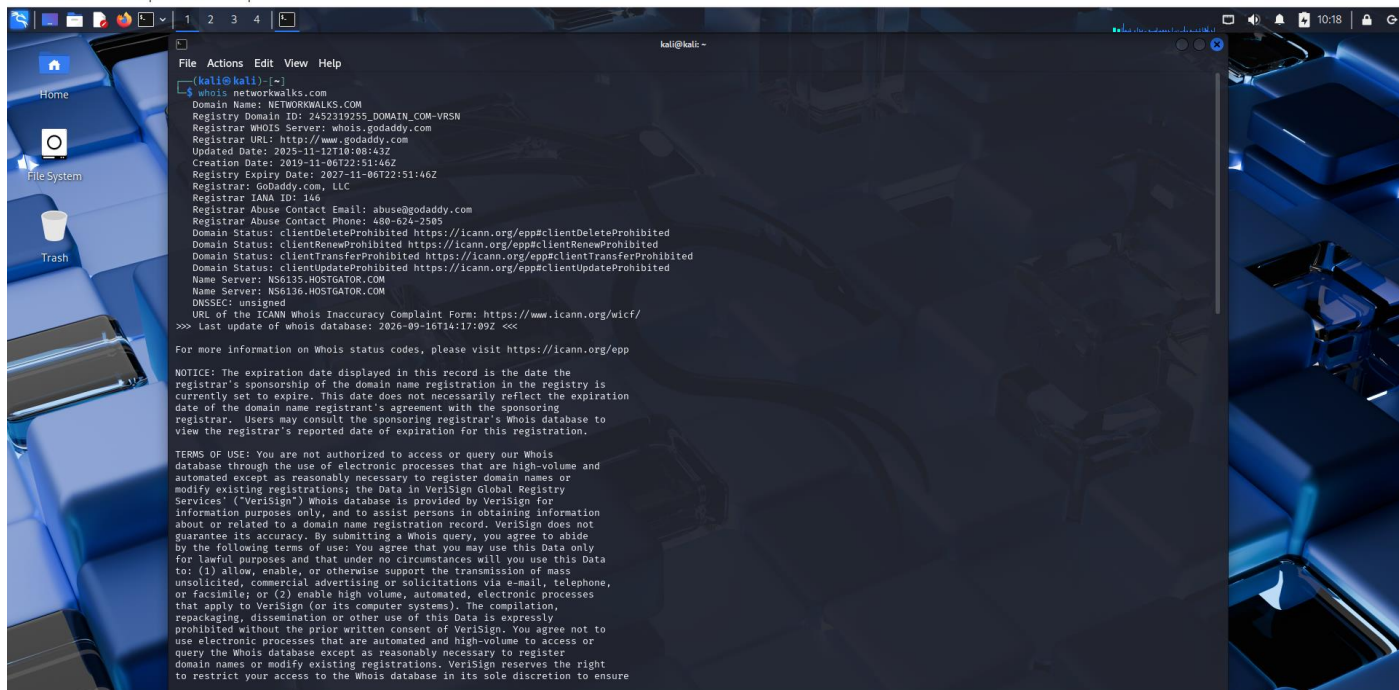
File Machine View Input Devices Help



EN USD/RUB +0.36% 7:16 PM 9/16/2026

kali-linux-2025.1c-virtualbox-amd64 (Fresh setup - 10.0.0.2) [Running] - Oracle VirtualBox

File Machine View Input Devices Help



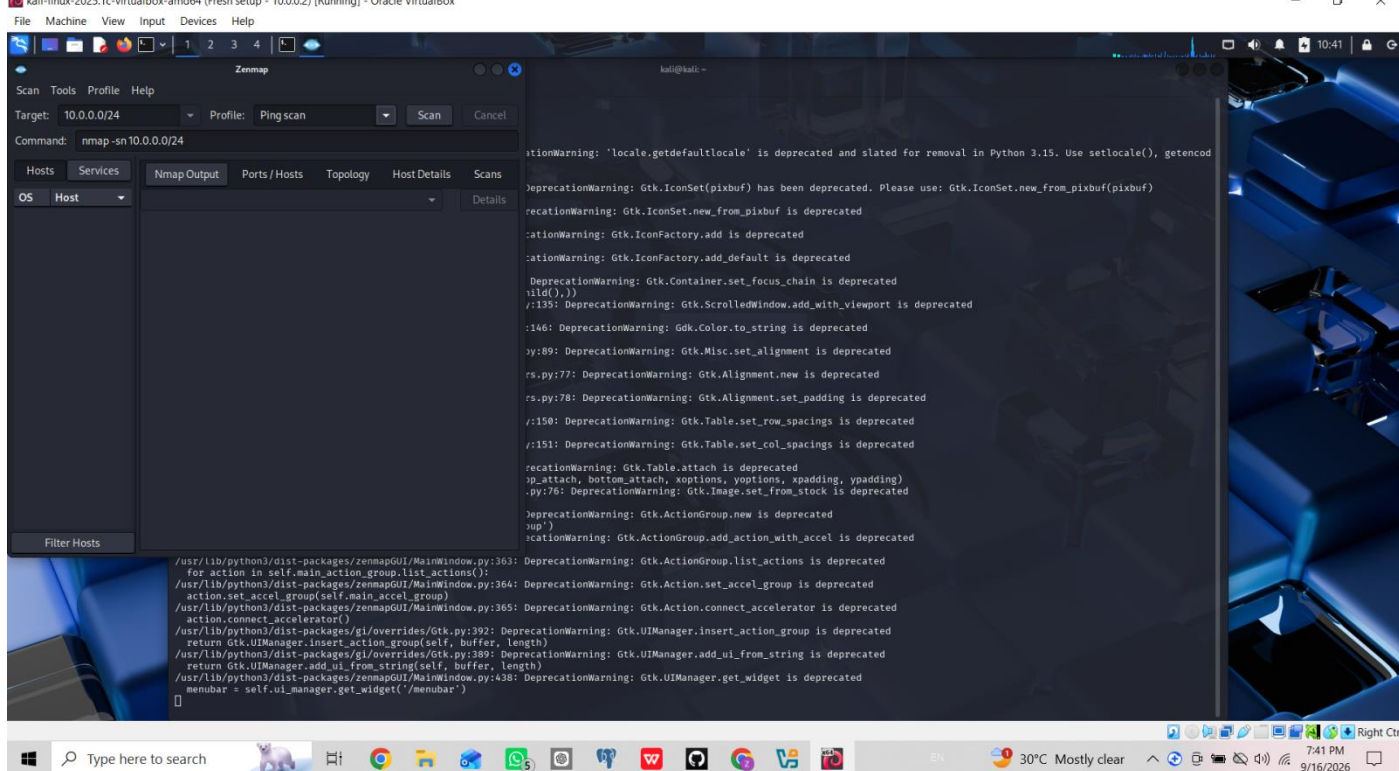
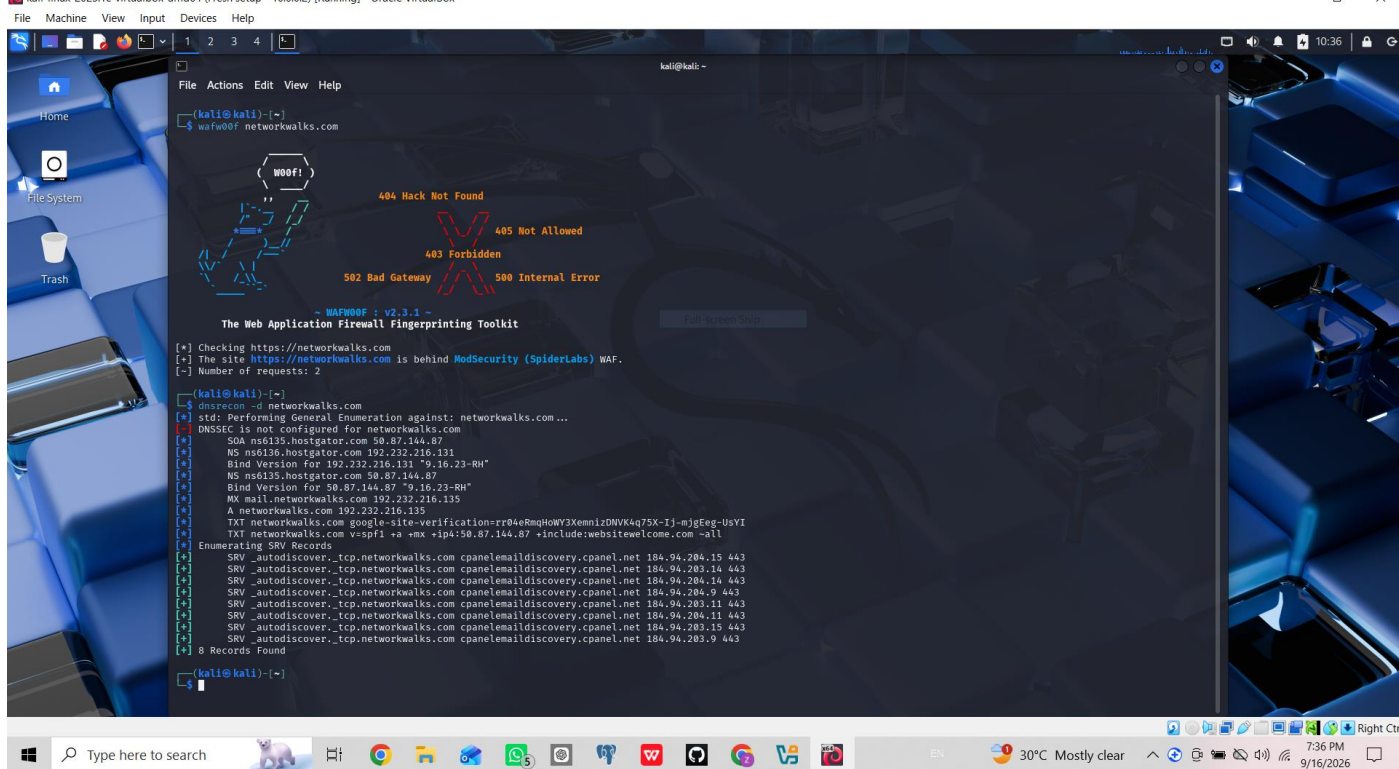
EN Rain tomorrow 7:18 PM 9/16/2026

```
kali@kali: ~  
$ whatweb networkwalks.com  
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[[_wpdm_client]], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]  
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[_wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IP1B&pid=nVar2=50511&prtVar2=76&pscVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]  
  
kali@kali: ~  
$
```

```
kali@kali: ~  
$ nslookup networkwalks.com  
Server:      8.8.8.8  
Address:     8.8.8.8#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135  
  
kali@kali: ~  
$
```

```
kali@kali: ~  
$ curl -I https://networkwalks.com  
HTTP/2 200  
permissions-policy: private-state-token-redemption(self="https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://h  
captcha.com"), private-state-token-issuance(self="https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcap  
tcha.com")  
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <h  
ttps://networkwalks.com/>; rel=shortlink  
set-cookie: wpda_client=bb6b448ed96881adeab8ec751f9c7fe; path=/; domain=networkwalks.com; secure; HttpOnly  
referrer-policy: no-referrer-when-downgrade  
x-endurance-cache-level: 0  
x-engine-cache: WordPress  
content-type: text/html; charset=UTF-8  
date: Wed, 16 Sep 2026 14:26:07 GMT  
server: Apache  
  
kali@kali: ~  
$  
kali@kali: ~  
$
```

```
kali@kali: ~  
$ wafw00f networkwalks.com  
  
{ W00f! }  
404 Hack Not Found  
405 Not Allowed  
403 Forbidden  
502 Bad Gateway  
500 Internal Error  
  
~ WAFW00F - v2.2.1 ~  
The Web Application Firewall Fingerprinting Toolkit  
[*] Checking https://networkwalks.com  
[*] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[*] Number of requests: 2  
  
kali@kali: ~  
$
```



kali-linux-2025.1c-virtualbox-amd64 (Fresh setup - 10.0.0.2) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

zenmap

Scan Tools Profile Help

Target: 10.0.0.0/24 Profile: Pingscan Scan Cancel

Command: nmap -sn 10.0.0.0/24

Hosts Services

OS Host

10.0.0.1

10.0.0.2

10.0.0.3

Starting Nmap 7.95 (<https://nmap.org>) at 2026-09-16 10:41 EDT

Nmap scan report for 10.0.0.1

Host is up (0.00077s latency).

MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)

Nmap scan report for 10.0.0.3

Host is up (0.0013s latency).

MAC Address: 00:00:27:02:A0:5A (PCS Systemtechnik/Oracle VirtualBox Virtual NIC)

Nmap scan report for 10.0.0.2

Host is up.

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.20 seconds

Filter Hosts

DeprecationWarning: 'locale.getdefaultlocale' is deprecated and slated for removal in Python 3.15. Use setlocale(), getencod

DeprecationWarning: Gtk.IconSet(pixbuf) has been deprecated. Please use: Gtk.IconSet.new_from_pixbuf(pixbuf)

reprecationWarning: Gtk.IconSet.new_from_pixbuf is deprecated

ationWarning: Gtk.IconFactory.add is deprecated

ationWarning: Gtk.IconFactory.add_default is deprecated

DeprecationWarning: Gtk.Container.set_focus_chain is deprecated

ild().))

/:135: DeprecationWarning: Gtk.ScrolledWindow.add_with_viewport is deprecated

/:146: DeprecationWarning: Gdk.Color.to_string is deprecated

py:89: DeprecationWarning: Gtk.Misc.set_alignment is deprecated

rs.py:77: DeprecationWarning: Gtk.Alignment.new is deprecated

rs.py:78: DeprecationWarning: Gtk.Alignment.set_padding is deprecated

/:150: DeprecationWarning: Gtk.Table.set_row_spacings is deprecated

/:151: DeprecationWarning: Gtk.Table.set_col_spacings is deprecated

reprecationWarning: Gtk.Table.attach is deprecated

ip_attach, bottom_attach, xoptions, yoptions, xpadding, ypadding)

.py:76: DeprecationWarning: Gtk.Image.set_from_stock is deprecated

DeprecationWarning: Gtk.ActionGroup.new is deprecated

map')

icationWarning: Gtk.ActionGroup.add_action_with_accel is deprecated

DeprecationWarning: Gtk.ActionGroup.list_actions is deprecated

/usr/lib/python3/dist-packages/zenmapGUI/MainWindow.py:364: DeprecationWarning: Gtk.Action.set_accel_group is deprecated

action.set_accel_group(self.main_accel_group)

/usr/lib/python3/dist-packages/zenmapGUI/MainWindow.py:365: DeprecationWarning: Gtk.Action.connect_accelerator is deprecated

action.connect_accelerator()

/usr/lib/python3/dist-packages/gi/overrides/gtk.py:392: DeprecationWarning: Gtk.UIManager.insert_action_group is deprecated

return Gtk.UIManager.insert_action_group(self, buffer, length)

/usr/lib/python3/dist-packages/gi/overrides/gtk.py:389: DeprecationWarning: Gtk.UIManager.add_ui_from_string is deprecated

return Gtk.UIManager.add_ui_from_string(self, buffer, length)

/usr/lib/python3/dist-packages/zenmapGUI/MainWindow.py:438: DeprecationWarning: Gtk.UIManager.get_widget is deprecated

menubar = self.ui_manager.get_widget('/menubar')

/usr/lib/python3/dist-packages/zenmapGUI/ScanHostDetailsPage.py:243: DeprecationWarning: Gtk.Table.set_col_spacings is deprecated

table.set_col_spacings(1, 50)

0

Type here to search

30°C Mostly clear

7:41 PM

9/16/2026

kali-linux-2025.1c-virtualbox-amd64 (Fresh setup - 10.0.0.2) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

zenmap

Scan Tools Profile Help

Target: 10.0.0.0/24 Profile: Pingscan Scan Cancel

Command: nmap -sn 10.0.0.0/24

Hosts Services

OS Host

10.0.0.1

10.0.0.2

10.0.0.3

Legend Save Graphic

Hosts Viewer Fisheye Controls

10.0.0.2

10.0.0.3

localhost

10.0.0.1

Filter Hosts

Type here to search

30°C Mostly clear

7:48 PM

9/16/2026

-End-

Thank you